Explainable Intrusion Detection in Industrial Control Systems

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***Abstract*— In the realm of smart manufacturing, Industrial Control Systems (ICSs) play a critical role in supervising essential operations across various manufacturing facilities. As ICSs become increasingly integral to critical infrastructure, their susceptibility to cyber threats escalates. Despite the positive impact of abundant digital data communication on ICS performance, rising concerns about data security underscore the pivotal role of intrusion detection systems (IDS) in averting the consequences of network security attacks. However, challenges persist, particularly in the detection of zero-day attacks. Furthermore, while many IDSs exhibit high performance for known attack patterns, they often function as black boxes, presenting interpretability challenges for network operators to take appropriate actions. This paper introduces a novel zero-day intrusion detection system, thoughtfully crafted to address the interdependence within ICS network traffic. Our approach designs a multi-head-attention mechanism, not only for precise classification of network traffic but also to mitigate zero-day attacks. Subsequently, we incorporate a novel autoencoder architecture, purposefully designed to capture distinctive patterns present in both known attacks and normal traffic, leveraging the output from the attention layer. Furthermore, our approach integrates Explainable Artificial Intelligence mechanisms, employing the multi-head attentions model to offer a more detailed description of the underlying process involved in traffic categorization. The model's efficacy is demonstrated through experiments conducted using water system ICS testbeds, underscoring its performance and reliability in detecting diverse cyber-attacks and inferring zero-day attacks with an accuracy surpassing 89%, while also providing interpretable results.**

Keywords: Industrial Control Systems, Zero-day attacks, Intrusion detection, SCADA systems.

I. Introduction

An Industrial Cyber Control System constitutes a comprehensive framework that seamlessly integrates hardware and software solutions, strategically devised to monitor, control, and secure industrial processes within a cyber-physical environment[1]. This system plays a pivotal role in the automation of processes across diverse industries, spanning energy, health systems and transportation. The core functions of an Industrial Control System (ICS) encompass supervisory control, data acquisition, and real-time monitoring of industrial operations[2], functioning as an intelligent manufacturing system overseeing diverse facilities through the integration of sensors and computing technologies.

Industrial Control Systems (ICSs) primarily consist of an Operational Technology network, which amalgamates both hardware and software components to oversee and control industrial equipment, assets, processes, and events. Key elements within this network include Programmable Logic Controllers (PLCs), sensors, and actuators [1]. Additionally, ICSs incorporate a traditional Information Technology network housing workstations, servers, and various computing devices.

Although Industrial Control Systems (ICS) traditionally operated in isolated environments with proprietary hardware and software, recent advancements in wireless communication have enabled ICSs to leverage Cloud Computing and data analytics, leading to improved overall performance. However, these advancements also raise the risk of cybersecurity vulnerabilities, potentially leading to severe physical and economic consequences [3].

In recent years, industrial systems have faced severe cyber threats, exemplified by the Stuxnet attack on Iranian uranium enrichment facilities[4][5], hacking attacks on the Ukrainian power grid, and malware attacks on a Saudi Arabian petrochemical plant[4]. These incidents, as demonstrated in the case of the fuel pipeline incident in the U.S., where fuel shortages along the East Coast persisted for several days, highlight the ease with which attackers can cause significant disruptions to the ICS[4]. By exploiting software vulnerabilities and disrupting network operations through the manipulation of communication data, attackers have the potential to cause extensive disruptions, especially when attacks extend to damaging the physical assets of the industrial facilities. As a result, safeguarding against unauthorized access, data breaches, and disruptions to critical infrastructure becomes imperative, particularly with the growing interconnectivity of industrial systems [3].

Tailored intrusion detection systems for Industrial Control Systems have demonstrated effectiveness in identifying known abnormal attack patterns. However, it is crucial to carefully consider the unique requirements of an ICS when selecting an intrusion detection solution that can effectively balance security and ensure uninterrupted services in ICS.

Achieving design goals for ICS faces the following challenges: (i) Ensuring continuous detection of emergent attacks by the intrusion detection system is imperative, demanding high sensitivity even for rare and zero-day attacks; (ii) Existing intrusion systems lack transparency in providing reasons for predicted results. This is critical as anomalies may not always stem from cyber-attacks but could originate from technical issues like faulty sensor readings, indicating potential failures[6][7]. It is worth noting that system faults involve internal malfunctions within the industrial control system, while network attacks encompass deliberate unauthorized actions targeting the communication infrastructure of the system for disruption, manipulation, or unauthorized access.

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Consequently, analyzing the cause of the decision-making of the IDS becomes important; (iii) The lack of transparency in existing IDS also introduces potential complexity for network operators in making informed decisions, as it adds complexity to decision analysis during anomalies.

In this paper, we employ a multi-head attention architecture, showcasing its exceptional effectiveness in identifying intrusions within dynamic industrial control system communications. Our approach incorporates an attention mechanism that not only allows the system to distinguish various network behaviors but also scrutinizes the influential features driving the predicted anomalies. This facilitates the identification of the causes associated with specific ICS abnormalities. By examining attention weights, we can validate whether anomalies result from actual cyberattacks, enabling swift responses and reducing decision-making time for the network administrators while bolstering IDS reliability. Furthermore, utilizing activation weights, we construct a lightweight autoencoder-decoder capable of discerning zero-day attacks from known attack patterns. Our distinctive contribution lies in the innovative integration of zero-day attack detection and explainability in our comprehensive Intrusion Detection System (IDS), enhancing anomaly interpretation and enabling network operators to efficiently mitigate attacks. The performance evaluation conducted on the Water System ICS dataset [8] affirms the efficacy of our approach. Our contributions can be summarized as follow:

- We propose a multi-head attention architecture capable of dynamically discerning the sequence of network flows displaying specific abnormalities.
- We design an autoencoder architecture guided by the output of the multi-attention mechanism, complemented by the confidence of the multi-head attention model in identifying abnormalities. This design aims to extract patterns linked to unknown abnormalities.
- We integrate the multi-head attention mechanism into explainable AI techniques to deliver interpretable results promptly, enhancing the decision-making process and improving the reliability of our IDS.
- We validated the effectiveness of our architecture in detecting diverse known and unknown attacks.

The paper is organized as follows. The next section covers the related work. Section III discusses the attack models. Section IV presents the intrusion detection System Details. Section V reports the validation results. Finally, the paper is concluded in Section VI.

II. Related Work

The vulnerability of ICS systems to cyber-attacks underscores the potential for severe repercussions, ranging from service disruptions to the compromise of sensitive information and physical damage to critical equipment [10]. Previous investigations into industrial intrusion techniques have broadly classified methodologies into signature-based [11] or anomaly detection-based approaches[13]. The incorporation of machine learning techniques for ICS security, as proposed in [12][14], focuses on detecting abnormalities using various algorithms such as support vector machines, and Naïve Bayes. Recent

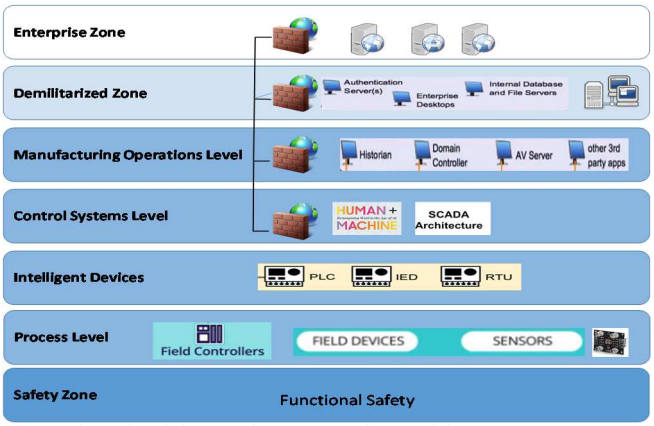


Fig. 1. The Industrial Control System Purdue Model.

research emphasizes the significance of deep learning in strengthening SCADA security, exploring diverse deep learning architectures to mitigate attacks in different ICS[15]. On another hand, Wang et al., [16], introduced a stacked deep learning-driven method for detecting cyber-attacks, demonstrating superior performance in distinguishing suspicious behaviors from normal actions compared to traditional models. The use of deep learning extends to convolutional neural networks (CNN) to detect cyber threats in SCADA systems [17]. These methods prove effective not only for normal and abnormal traffic detection but also for identifying various cyber-attacks, including distributed denial-of-service attacks [17]. Kim et al. [18] focused on exploiting different features like payload-based anomaly detection, while Canizo et al. [19] proposed a supervised multi-time-series anomaly detection method. Feng et al. [20] propose an anomaly detection method combining network packet content analysis and time-series structure analysis, utilizing a baseline signature database for common packets to train a stacked LSTM network and predict the most likely next packet signature. Ding et al. [21] integrate Long-Short Term Memory (LSTM) with Gaussian Mixture Model to balance accuracy and real-time requirements simultaneously. Graph Neural Networks (GNN) for anomaly detection have also been considered to capture underlying relationships. While these approaches introduce complexity, they aim to achieve high accuracy and adeptly address emerging threats[22]. Yet, the mentioned methodologies have not addressed concerns related to the resilience of the model against unknown attacks and the creation of interpretable attack detection.

In addition, the authors [23]introduced an Intrusion Detection Method based on Hybrid Machine Learning in the Industrial Control Field to address data imbalance in industrial control system using Transformer modules and BiGRU, facilitating the discovery of internal relationships between features. However, these approaches do not account for zero-day attacks and lack explainability. Graph Attention has also been employed to detect hidden advanced persistent threats in the I-IoT-enabled Cyber Physical System[24]. This approach utilizes masked self-attentional layers to overcome limitations in prior Deep Learning but introduces excessive model complexity. Finally, while the authors [25]proposed Explainable Artificial

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Intelligence to enhance the reliability of the LSTM-based Autoencoder-OCSVM learning model for anomaly detection in ICS, their approach does not consider the presence of zero-day attacks. To our knowledge, this is the first IDS for ICS that comprehensively exploits the self-attention mechanism for the detection of diverse known and unknown zero-day attacks while providing comprehensive explainability through an exact mapping of features and attention weights with less complexity.

III. Attack Model and Background

A. Attack Model

The vulnerabilities within ICS protocols stem from the absence of effective authentication and communication integrity mechanisms. These shortcomings create openings for attackers to eavesdrop on communication links or manipulate network transmissions, posing a potential threat to ICS operations. Specifically, attackers can execute various attacks, ranging from network discovery to packet sniffing or modification, ultimately disrupting ICS functionality [27].

Reconnaissance *Attack:* It is a passive attack wherein the attacker gathers crucial system information to subsequently launch more advanced attacks without causing disruption to the victim's functions. IP scanning and port scanning are potential methods for executing such attacks.

Replay Attack: The attacker exploits valid network packets captured during regular ICS operations to maliciously retransmit them later, disrupting the system's normal behavior. Replaying such packets at inappropriate times can lead to unexpected results; this attack doesn't require in-depth knowledge of network traffic packets.

Distributed Denial of Service (DDoS): Attackers flood the network with a large volume of packets, leading to a denial of service on the ICS components. For instance, DDoS attackers can generate an extensive number of reading requests to the PLC, resulting in significant delays in ICS network communication [27].

Man-in-the-Middle Attack: The attacker injects false data into the control system, introducing inaccurate information that jeopardizes communication in ICSs. For example, the attacker intercepts packets, alters Modbus responses, and then dispatches the manipulated packets to the intended destination[27].

B. Industrial Control Systems Architecture

Industrial Control Systems comprise interconnected electrical and mechanical devices that oversee physical processes [1]. These systems integrate various hardware and software components, including sensors, actuators, physical systems, computational nodes, communication protocols, SCADA systems, and controllers [9]. Control within these systems can be automated or involve human interaction through a Human Machine Interface (HMI) [9]. In contrast to conventional networking systems, ICSs utilize standard TCP/IP network traffic to manage data from physical processes and low-level components within real-time constraints. This section offers a brief overview of the most common devices in the ICS systems and subsequently describes their integration with the ICS architecture[9]:

- The Programmable Logic Controllers (PLCs) functions as a microprocessor controlled electronic devices tasked with reading signals from sensors, executing instructions, and generating output signals to the actuators. Equipped with a Real-Time Operations System (RTOS), it ensures precise and timely execution of operations. The PLC comprises vital components, including a power supply, central processing unit, communication interfaces, and input-output modules[9].
- The Remote Terminal Unit (RTU) serves as a microprocessor controlled electronic device, featuring two distinct types: field RTUs, tasked with collecting data from field devices, and station RTUs, which receive data from field RTUs and relay orders from supervisory controllers. RTUs employ Wide Area Network technologies for communication with the control center.
- Front End Processor (FEP) serves as a communication processor, systematically collecting status information from diverse devices to monitor the overall system status.
- The Intelligent Electronic Device (IED) includes processors with the capability to receive or send data, fulfilling roles such as protection and control within substations. Instances of IEDs include digital relays and controllers, providing various functionalities, including fault detection.
- The Engineering Workstation, usually comprising computers or servers, is employed for overseeing controllers and applications, aiding in engineering tasks. The Human Machine Interface (HMI) is software installed on different devices, allowing operators to monitor and control automation processes.
- Communication Gateways play a vital role in translating packets between diverse protocols, ensuring smooth communication in industrial networks that utilize protocols like Modbus and DNP3[9].
- The Master Terminal Unit (MTU) manages bidirectional communication with PLCs, gathering and processing data.

As illustrated in Figure 1, the architecture of Industrial Control Systems (ICS) is systematically structured into istinct levels in the Purdue model[9]. The upper layer relates to control and monitoring applications. It includes IT devices and systems, such as logistic business systems and the network infrastructure [9]. The second layer, the Demilitarized Zone (DMZ), serves as a crucial intermediary controlling data exchange between the Control Zone and the Enterprise Zone, ensuring a secure and controlled linkage between the Information Technology (IT) and Operational Technology (OT) networks[9]. The Control Zone, is further divided into four hierarchical levels. At Level 0, sensors and actuators directly interact with the physical processes, while Level 1 accommodates intelligent devices like Programmable Logic Controllers (PLCs), IIEDs, and RTUs.

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Level 2 encompasses control systems, including HMIs, alarms, and control room workstations. Finally, at Level 3, manufacturing operation systems take charge to produce the desired final product. Finally, the last layer, the Safety Zone, is dedicated to ICS security management through anomalies and intrusion detection to prevent potential attacks.

IV. Approach Description

The critical nature of ICS systems, coupled with their remote connectivity, exposes vulnerabilities to a myriad of attacks across various hierarchy levels. These attacks leverage the networking and communication infrastructure, ranging from network scanning to sophisticated distributed denial of service or man-in-the-middle attacks. In response, machine learning (ML) and deep learning in particular, have emerged as successful tools for detecting malicious activities within networks. ML based Intrusion Detection Systems (IDSs) demonstrate generalization capabilities, identifying previously unseen forms of malicious data that share similar patterns. Network traffic, being interdependent, makes sequence-based analysis particularly appealing for IDS in network communication. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) models have been widely employed for network flow analysis in diverse ICS networking infrastructures. However, challenges arise with increasing attack diversity and imbalanced training data. Data imbalance during training can lead models to prioritize high-instance classes, while overlooking low-instance classes and limiting the identification of rarely seen attacks. Furthermore, existing research predominantly focuses on identifying known attacks, neglecting the potential harm posed by zero-day attacks to the ICS. Our aim is not only to differentiate anomalies and identify various cyber-attacks but also to leverage a well-designed learning architecture to understand the behaviors of known attackers and offer explainable predictions.

To address the shortcomings of existing work, this paper employs a Multi-head Attention model to analyze ICS systems against a diverse range of cyberattacks. Multi-Head Attention utilizes multiple heads on sequences of network traffic to extract correlations over different timesteps, facilitating the dynamic learning of the most relevant features. Unlike other models, multi-head attention allows the selection of dominating features based on input values dynamically. More importantly, the self-attention mechanism within the Multi-Head Attention model enables focus on different parts of the input, enhancing the model's capabilities in rare attack scenarios. Additionally, we utilize the attention weights to detect zero-day attacks and offer comprehensive explanations for our IDS. Specifically, we design a lightweight autoencoder-decoder model fed with attention weights to detect unknown behavior and provide explanations for the detected attacks. A deep analysis of the outcomes derived from the dynamic self-attention mechanism enables us to discern patterns of known attacks. This is achieved by assessing the cosine similarity between the reconstructed weights obtained from the decoder and the attention layer weights. Consequently, our model captures dynamic network traffic properties, establishing a comprehensive framework for detecting both known and unknown attacks while offering explainable decisions for ICS administrators.

A. Cyber-Attacks Detection Engine

During ICS operations, a substantial volume of intricately interconnected information is exchanged. A thorough analysis of this data involves delving into feature sequences intricately linked to contextual flows. As a result, ICS data not only demonstrate essential temporal correlations but also encompass diverse levels of feature importance contingent on the analyzed data. Traditional approaches primarily hinge on temporal correlation, often neglecting dynamic feature selection based on the input and the utilization of dynamic feature selection to recognize known attack scenarios and detect unknown attack patterns. This limitation constrains their adaptability to cope with the dynamic nature of cyberattacks in ICS.

Acknowledging the interconnectivity and interdependence inherent in data transmitted across an ICS network, our approach shifts from scrutinizing individual flows to examining sequences of flow, utilizing a window of size ξ. More precisely, we extract λ features characterizing the network flow from incoming data packets, as outlined in the experiment section. Instead of relying on traditional machine learning models to independently infer the categorization of traffic, where feature importance is input independent, we apply a multi-head attention mechanism to dynamically select which features receive heightened attention. This mechanism equips the model with the ability to simultaneously focus on various segments of the input sequence, considering combinations of input features. In other words, the attention enhances the model's capacity to grasp intricate relationships and dependencies within the input data dynamically.

Our goal is to employ a self-attention mechanism [26] that enables any subsequence of the input to attend to any other subsequence and determine the most relevant features for any flow of packets. To achieve this, we implement a multi-head

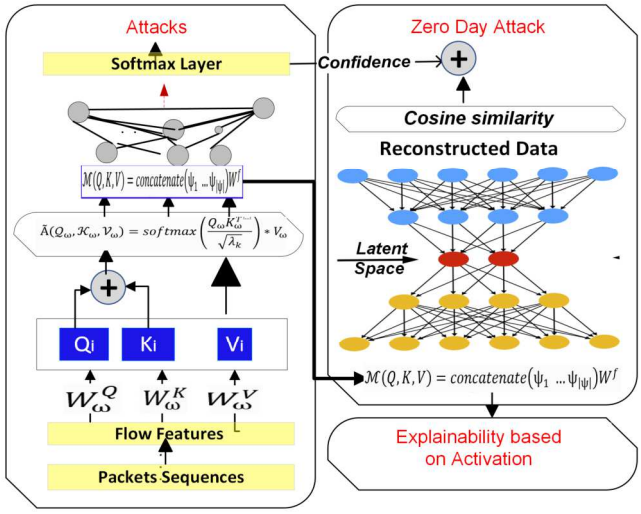


Fig. 2. The considered IDS architecture processes network flows in sequences to extract attention weights and predict the confidence vectors. These vectors are utilized by the autoencoder and decoder to estimate cosine similarities, subsequently aggregating knowledge to detect zero-day attacks and provide effective explainability.

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self-attention mechanism in which the input sequence vector is partitioned into smaller vectors, allowing the attention layer to focus on different aspects simultaneously. Each subsequence contributes a key, value, and query. The multi-head attention layer utilizes these three vectors - queries (Q), keys (K), and values (V) - to transform input vectors, extracting distinct aspects of the input data. The query in the context of multi-head attention is a feature vector that outlines the specific aspects we aim to focus on within the sequence, indicating which part in the sequence might be relevant for attention.

Each input element is connected to a key, forming a feature vector that defines the element's contribution. Additionally, for every input subsequence, there is a corresponding value vector, representing the feature vector over which we aim to compute an average[26]. The attention mechanism dynamically calculates weights based on input queries and keys. The comparison of queries with the actual value, using a scoring function namely the dot product of matrices, determines the weights. This dynamic weighting, depending on the actual values, allows us to discern which inputs are pertinent to the prediction. Subsequently, the value vectors are averaged using the acquired attention weights. This process results in a unique value vector for each head.

More formally, the Multi-Head Attention layer systematically handles a set of queries (Q with dimensions ξ×λ), keys (K with dimensions ξ×λ), and values (V with dimensions ξ×λ), where ξ represents the number of packet flows in a sequence, and λ represents the number of flow features. Within each head (denoted as ω in a set of ψ heads) in our self-attention layer, the queries Q, keys K, and values V undergo linear projection as follows [26]:

$Q_\omega = QW_\omega^Q$, $K_h = KW_\omega^K$, and $V_\omega = VW_\omega^V$, where $W_\omega^Q \in \xi \times \lambda_k$, $W_\omega^K \in \lambda \times \lambda_k$, and $W_\omega^V \in \lambda \times \lambda_v$ are learned weight matrices. Given the difference in the learned weights $W_\omega^K, W_\omega^Q, and\ W_\omega^V$, the projections will enable the self-attention mechanism to learn different information from the sequence of flows in the aggregated packets. In accordance with the recommended dimensions for queries and values [26], we assume that the projected queries and keys have a size of λ_k, while the projected values have a size of $\lambda_v = \lfloor \lambda / |\psi| \rfloor$.

To determine the elements deserving attention, a score function is utilized, taking the obtained Q_ω, V_ω and K_ω as inputs and producing the score or attention weight. This function involves simple similarity metrics using a dot product. The matrix multiplication $Q_\omega K_\omega^T$ performs the dot product for every pair of queries and keys. Each row represents the attention logits for a specific element of the input to all other elements in the sequence. The scaling factor of $1/\sqrt{\lambda_k}$ is crucial to maintain appropriate variance in the attention values. The weights are then derived through a softmax operation, assigning higher weights to value vectors whose corresponding keys exhibit greater similarity to the query. The scaled dot-product attention mechanism Ã can be expressed as follow:

$$\tilde{A}(\mathcal{Q}_\omega, \mathcal{K}_\omega, \mathcal{V}_\omega) = softmax\left(\frac{Q_\omega K_\omega^T}{\sqrt{\lambda_k}}\right) * V_\omega$$

Table 1:. The size of the training and testing datasets.

Label	# Training Samples	# Test Samples
0-Normal	24559	12146
1-IP-Scan	137	55
2-Replay	1590	766
3- port scan	1317	627
4-DDOS	1326	608
5-man in the middle	1700	884

To empower the model to focus on various aspects of the input, the attention results of each head are concatenated and integrated with a final weight matrix W^f. In essence, all heads are consolidated and subject to linear projection using the weight matrix W^f, The concatenation of results is accomplished as follows to form the definitive output of our Multi-Head Attention layer $\mathcal{M}$:

$$\mathcal{M}(Q,K,V) = concatenate(\psi_1 \ldots \psi_{|\psi|})W^f$$

$$\psi_\omega = \tilde{A}(\mathcal{Q}_\omega, \mathcal{K}_\omega, \mathcal{V}_\omega)$$

Following this, we utilize two dense layers linked to a softmax layer with N outputs, where N signifies the potential known normal and abnormal behaviors. The network is fine-tuned using cross-entropy loss for intrusion Detection. Additionally, the model generates attention weights for each input during testing, which are employed for explainability and zero-day attack detection, as further elaborated in the following subsections.

B. Zero-day Attack Detection Engine

The multi-head attention model, effective in addressing known attacks during training, encounters challenges with zero-day attacks and unfamiliar patterns. In this paper, we leverage attention weights to analyze potential discrepancies in the model's behavior when categorizing traffic flows, specifically zero-day attacks, in contrast to known attacks. However, the prevalence of normal data, representing the majority class in regular ICS traffic, poses a dilemma for supervised models, in general, resulting in the misclassification of unknown attacks as normal. This issue is inherent in supervised learning models, where the model tends to predict unknown behavior as benign. To address this limitation in existing supervised learning, we employ two lightweights deep autoencoders-decoders to establish a mechanism that distinguishes benign traffic, known malicious attacks, and zero-day attacks. The attention layer output directly feeds both encoders, facilitating the learning of dynamic feature representations. More precisely, we rely on the fact that the attention weights offer high interpretability, highlighting the specific focus of each head in sequences, particularly in consecutive network flows.

Specifically, we design a lightweight autoencoder architecture featuring 5 layers with 64, 32, 16, and 8 neurons and Rectified Linear Unit (ReLU) activation functions as well as a 0.2 dropout hidden layer. Correspondingly, the decoder is configured with layers consisting of 8, 32, 16 and ξ×λ neurons, each employing the ReLU activation function. The training process is facilitated by the Adam optimizer and the cosine similarity loss function. The primary goal is to minimize the disparity between the embedded sequence of attention weights and the decoded vector

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while preserving the extracted features' importance dynamically captured by the previously described multi-head attention architecture. To ensure unbiased output, we have incorporated a Normalization layer into our autoencoder-decoder.

By placing significant emphasis on attention during training, the autoencoder progressively refines the learning process for both known malicious and normal traffic, optimizing the cosine similarity of the attention weights and the reconstructed vector of weights. In testing, the computation of cosine similarity is employed to discern the similarity of the decoded attention weights, and the results are juxtaposed against the threshold ¥ observed during training to ascertain whether the patterns are indicative of known or unknown attacks. Moreover, in instances where the testing cosine similarity surpasses the threshold ¥, our engine integrates the classifier's confidence to discern unknown patterns while minimizing false positives.

To be more specific, we extract sequences of packet flows that not only exhibit lower similarity than the threshold but also demonstrate low confidence in predictions, even if the similarity metric exceeds the threshold. To gauge confidence, we leverage the probability distribution from our IDS engine. Notably, the output of our model incorporates a softmax layer, offering a probability distribution for the N possible classes of attacks and normal traffic. When the cosine falls below the threshold, the flow is deemed suspicious, especially when confidence is low in detecting specific malicious behavior, signifying a heightened risk of a zero-day attack. Conversely, if confidence is high in detecting a specific behavior and the cosine surpasses the threshold, the flow is identified as a known behavior. In both scenarios, we turn to explainability to evaluate the flow. We create a Shapley features importance map for each attack distribution, enabling a more nuanced comparison of the flow's characteristics with the flagged behavior determined by our autoencoder-decoder. The security analyst utilizes all these insights to determine the appropriate action to address zero-day attacks. Figure 2 illustrates the overall architecture of our system.

V. Validation Experiments

Our approach's efficacy is substantiated through validation on the water ICS dataset [27], encompassing flow features defined as aggregated network packets within a specific time interval (500 ms), characterized by similar source/destination addresses and network protocols. Table 1 presents statistics for the dataset used in both training and testing. The dataset originates from a testbed simulating a bottle-filling factory [27], where the ICS oversees devices such as pipes, valves, a conveyor belt, and a water tank [27]. The ICS's goal is to fill empty bottles with water sourced from the tank. The input valve maintains the water level within the permissible range, and the output valve regulates water flow for bottle filling. PLC-1 reads sensors measuring tank water level and pipe water flow, issuing control commands to activate or deactivate input and output valves. PLC-2 monitors water levels in filling bottles and the spacing between the filler and the next bottle using sensors, directing the conveyor belt based on sensor readings and communication with PLC-1. The conveyor belt engine ensures proper alignment of empty bottles beneath the filler[27]. PLCs function as Modbus-TCP servers, with other components acting as Modbus-TCP clients. Leveraging this dataset [27], our focus is on detecting potential attackers remotely gaining access to the ICS or breaching the ICS internal network through compromised remote HMIs. We utilize the network flow features detailed in [27] to detect known and unknown attack patterns.

We first construct the attention layers using 6 keys and 6 heads and connect this layer with the multilayer neural network architecture comprising two hidden layers with 64 neurons each. Activation is achieved through Rectified Linear Units (ReLU), and the final layer employs SoftMax with N=6 outputs, representing the attack classes considered by the IDS in our implementation. This configuration is selected given the evaluation results which provide satisfactory attacks detection with less complexity. In assessing the effectiveness of our solution and, consequently, the detectability of attacks, we conduct a performance comparison with baseline approach [27] that utilizes an identical prototype, ensuring a fair comparison of performance. This comparative analysis offers insights into the solution's performance in detecting diverse cyberattacks. The performance metrics include:

- Accuracy F1-score, recall, and precision of the solution in identifying cyberattacks compared to baseline models.
- Accuracy, F1-score, recall, and precision for detecting zero-day attacks.

A. Performance Results

Table 2 illustrates the superior performance of our approach compared to various ML classifiers proposed in the baseline approach using same dataset [27]. Table 2 demonstrates that our solution surpasses all benchmark models proposed in the baseline approach in terms of accuracy, recall, F1 score and precision. The results are attributed to the advanced neural network architecture employs.

In Figure 3, we showcase the efficacy of our zero-day attack detection mechanism, demonstrating a remarkable level of accuracy even when we varied the category of unknown attacks. Essentially, we randomly selected a specific malicious behavior to eliminate from the training, evaluating the ability to detect these unknown attacks using our integrated autoencoder and decoder using the outcome of the attention mechanism. As we varied the type of attacks (classes), our results remained consistent, affirming the approach capacity to distinguish between known and unknown behaviors. The congruence between zero-day detection performance and correct classification reinforces the reliability of our solution, as indicated by the high precision, recall, and F1-score metrics provided in Figure 3.

The robust performance of our system is attributed to the dynamic determination of attention weights embedded into encoder layers, enabling meticulous discernment of known patterns. The dynamic attention weights in the detection procedure play a crucial role in recognizing both known and unknown attack patterns. Indeed, in Figure 4, the cosine similarity plot illustrates a substantial overlap between training and testing errors distribution, while known malicious and

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unknown behaviors are displaying distinct distributions. To enhance the accuracy of detecting unknown attacks, we analyze both the model prediction confidence and the cosine error results. Precise classification of zero-day attacks is achieved by identifying samples with low cosine similarity (¥<0.85), obtained by running the encoder-decoder output against the input attentions, coupled with lower confidence (€<0.9) from our model predictions. The threshold for the cosine similarity and the confidence are extracted according to the validation results. The ability of our approach to offer effective explanations and prompt interpretation of results is further validated in Figure 5-6, illustrating distinctive Shapley values for each attack type with high precision. The explanatory analysis proves valuable in identifying if the flow aligns with a specific attack pattern, utilizing feature importance as depicted in Figure 5-6. A distinctive aspect of our model lies in establishing an equivalent mapping between the original features and the derived attention output. From Fig. 5, it is evident that features such as the maximum interval between packets and their acknowledgment in the receiver, along with the average inter-packet interval, average delay for acknowledgments, and the bit rate of sent data per second, contribute to the understanding of normal pattern. The comprehensive list of features is elucidated in [27]. The numbers at the end of each label indicate the rank of the features relative to the end of the sequence length.

Similarly, we can compute the Shapley value for additional attack samples. The significance of these features empowers the network operator to meticulously validate the identified features from our system, ensuring the authenticity of the alerts we generate. The precision with which our system accurately distinguishes similarities within the feature space is attributed to the meticulous mapping between attention weights and flow sequences. For example, the number of bytes sent or received as a payload in one packet within the flow, the rate of synchronization packets, the average interval between packets and their acknowledgment in the sender or receiver, as well as the maximum interval between packets and their acknowledgment in either the sender or receiver node, collectively provide valuable information for identifying DDoS threats. Monitoring these parameters collectively strengthens the network's resilience against DDoS attacks and other cyberattacks enabling reactive measures and timely intervention. Likewise, we have the capability to generate explanations for all other types of attacks. It is worth noting that these results enable network administrators to determine appropriate matching for the attacks using this comprehensive framework.

V. Conclusions

This paper has presented a novel intrusion detection for ICS to effectively counter zero-day attacks. It underscores the crucial need for innovative solutions that not only adeptly handle known attack patterns but are also finely tuned for the real-time dynamics of Industrial Control Systems (ICS) and Supervisory Control and Data Acquisition (SCADA) systems. Our approach employs Multi-Head Attentions, intricately designed to dynamically capture inherent correlations in ICS traffic. Going beyond conventional feature extraction, this model utilizes an attention-based mechanism for precise network traffic classification, even in the presence of rare and

Attack	Accuracy				Precision			
	Approach	DT	RF	ANN	Approach	DT	RF	ANN
0	.99	.99	.99	.99	1.	.99	.99	.99
1	.99	.99	.99	.99	1.	1.	1.	.99
2	.93	.92	.92	.89	.46	.35	.36	.36
3	.95	.93	.95	.94	.93	.90	.91	.90
4	.97	.95	.98	.95	.98	.94	.94	.93
5	.95	.9	.97	.90	.97	.93	.95	.98
	Recall				F1-Score			
	Approach	DT	RF	ANN	Approach	DT	RF	ANN
0	1.	.99	.99	.99	1	.99	.99	.99
1	.99	.99	.99	.99	1	.99	.99	.99
2	.93	.94	.98	.56	.62	.52	.53	.46
3	.94	.88	.90	.92	.94	.89	.90	.91
4	.96	.93	.93	.89	.97	.94	.94	.91
5	.90	.90	.89	.88	.93	.92	.92	.93

Table 2: Comparing the accuracy, recall, and precision of cyberattack detection between our model and three baseline models.

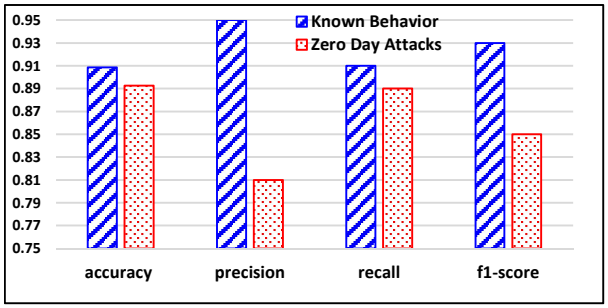


Figure 3: Comparing the accuracy, recall, precision, and F1-score of known cyberattacks and zero-day attacks detection using the autoencoder engine.

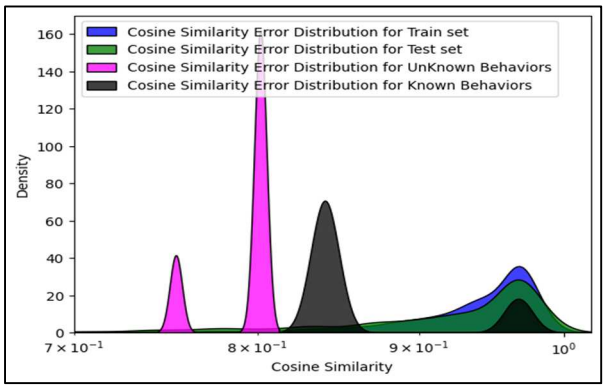


Figure 4: Comparing the training and testing cosine similarities for known benign/malicious traffic and unknown behaviors.

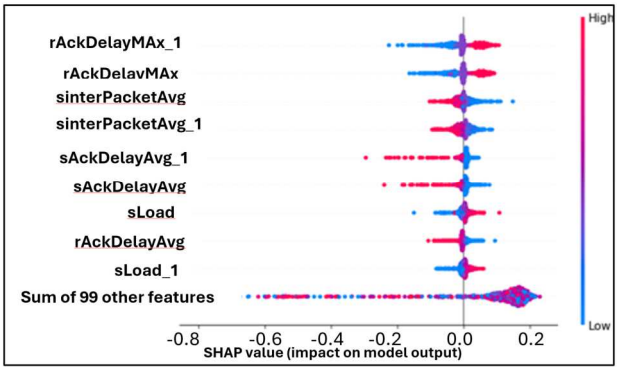


Figure 5: The Shapley feature importance for regular network traffic.

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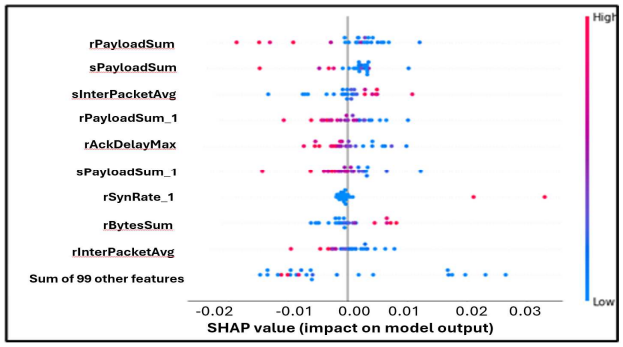


Figure 6: The Shapley feature importance for distributed DoS traffic.

sophisticated attacks. Furthermore, we integrate the Multi-Head Attentions model with Explainable Artificial Intelligence (AI), offering valuable insights into cyber-attacks detection and significantly enhancing risk assessment in ICS. To effectively address the challenge posed by zero-day attacks, we leverage the attention layer output to feed into a novel autoencoder-decoder. This autoencoder-decoder is designed to capture regular patterns of known attacks and normal traffic, thus enabling the detection of any zero-day attack within ICS. The experimental validation, conducted on water ICS testbeds, confirms the model's efficacy, establishing its robust performance and reliability in industrial settings.

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